

- 2 The structural protein collagen is the main component of the extracellular matrix (ECM) of the cells of the gas exchange system. The ECM is the supporting external environment of cells.

In the gas exchange system, the two main cell types that produce and secrete collagen are fibroblasts and chondrocytes.

- (a) Fibroblasts are found throughout the gas exchange system, but chondrocytes are localised within a particular type of tissue found in the trachea and in the bronchi.

- (i) Name the tissue located in the trachea and bronchi that contains chondrocytes.

..... [1]

- (ii) Describe features of the tissue named in **2(a)(i)** and outline the role of this tissue in the trachea and bronchi.

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..... [3]

- (b) In certain conditions, fibroblasts divide by mitosis to produce more cells. During this time, there are increased levels of an enzyme that acts to maintain the length of the telomeres.

Explain why maintaining telomere length is important for fibroblasts that carry out mitosis.

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..... [2]

- (c) Name the cell structures involved in the formation of the mitotic spindle in a fibroblast that are **not** present for the formation of the mitotic spindle in a plant cell.

..... [1]

- (d) The structure of a chromosome changes during mitosis.

Describe the structure of a chromosome as it appears at **telophase** of mitosis.

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..... [2]

- (e) Figure 2.1 shows part of an alveolus and an alveolar capillary separated by a very thin ECM space known as the interstitium.

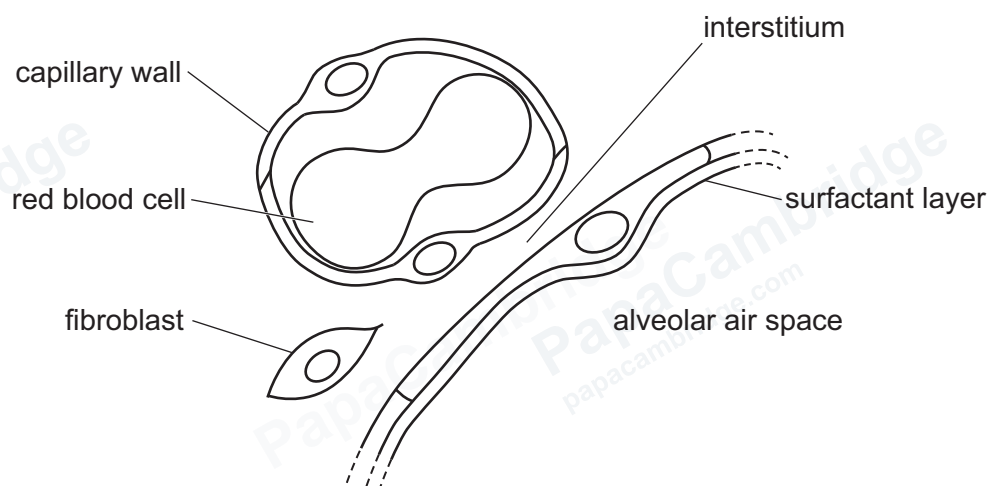


Figure 2.1

- (i) Name the type of cell shown in Figure 2.1 that forms the capillary wall.

..... [1]

- (ii) When damage occurs to the alveolar wall or capillary wall, fibroblasts become more active and increase the production of ECM components.

Suggest the change that will occur to the interstitium owing to an increased fibroblast activity **and** explain how this will result in less oxygen being transported away from the capillary.

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..... [2]

- (f) A collagen molecule is composed of more than one collagen polypeptide.

The primary structure of the collagen molecule contains a repeat sequence of three amino acid residues (amino acid triplet). Each amino acid triplet contains glycine (gly), as shown in Figure 2.2.

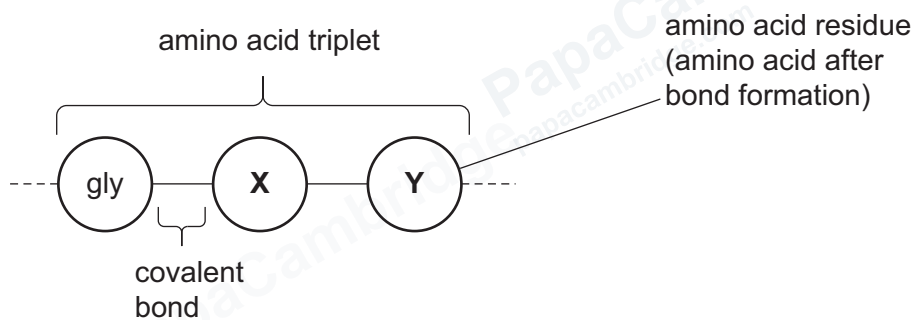


Figure 2.2

- (i) Name the type of covalent bond that links the amino acid residues shown in Figure 2.2.

..... [1]

- (ii) Three different amino acids are commonly found in the **X** and **Y** positions of the amino acid triplet. One of these amino acids, alanine, has a similar role to glycine in the structure of a collagen molecule.

The molecular structure of alanine is shown in Figure 2.3.

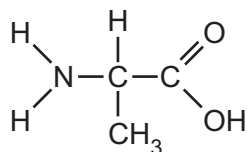


Figure 2.3

With reference to Figure 2.2 and Figure 2.3, explain why a high proportion of glycine and alanine residues is important for the structure of a collagen molecule.

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..... [3]

[Total: 16]